Lumbar CT segmentation and vertebral body centres

A. Preparing the data

I used the official TotalSegmentator v2.0.1 split: 1,082 training, 57 validation and 89 test scans. Training first used 32 scans. Later training used 64 scans in total: the original 32 plus 16 full-lumbar and 16 L1/L2-only scans. Final model selection used all 57 validation scans; testing used all 89 source-test and 73 VerSe 2020 test scans. VerSe labels 20–24 map to L1–L5 (output labels 1–5). L6 (label 25) is treated as background. Predicting it as L1–L5 counts as an error.

Both datasets were resampled to a 2 mm RAS grid (right, anterior, superior), using linear interpolation for CT and nearest neighbours for labels. This saves memory but loses small details. CT intensities were normalized using fixed statistics from training data.

Header issue (occurrences)	Detection	Handling
Missing units (7,450)	Missing spatial code in `xyzt_units`	Some files did not specify their measurement units. I treated the coordinates as millimetres, following the dataset's guidance, and checked that the CT scans and their label masks lined up.
CT/label grid mismatch (963)	Compared shapes and affine-transformed corners	The six cases with larger differences were inspected visually. No gross alignment error was observed.
Nonorthogonal axes (238)	Tested normalized affine-axis dot products	Used the original image coordinates when resampling.
Spacing conflict (5)	Compared `pixdim` with affine-axis lengths	Used the spacing from the image coordinates. The difference over the full scan was less than 0.1 mm.

I visually checked 24 cases and excluded seven training scans with overlapping labels. All test cases were kept and scored against their original labels. The supporting files contain the detailed checks.

B. Training and choosing the model

I used **nnU-Net v2 ResEnc M**, trained from scratch, as an established 3D baseline that fit the compute budget. Training used 128^3 voxel patches, batch size 2 and SGD. I used intensity augmentation and sampled partial L2 examples more often. The loss combined Dice with cross-entropy on the hardest 10% of voxels.

I selected **epoch 15** using whole-scan validation. Its mean Dice was **0.826**, with no predictions on the 17 scans without lumbar labels. Epoch 20 scored slightly higher (0.834) but produced false positives, so I kept epoch 15.

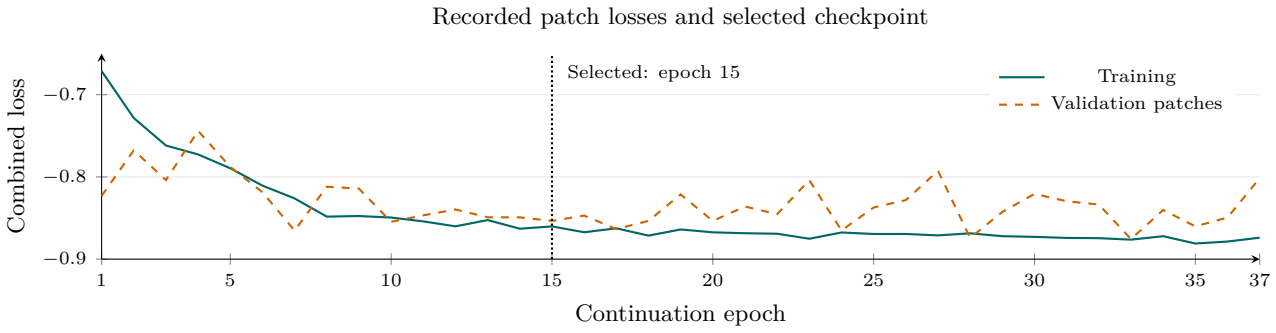


Figure 1. Loss is Top-K cross-entropy plus negative soft Dice, so negative totals are expected. Patch loss was not the selection metric. Learning rate decayed from 0.001 (polynomial power 0.9); after epoch 10 its horizon was extended to epoch 100 without raising the rate. Later epochs are outside the selected checkpoint.

C. Finding the vertebral body centre

I estimated **570 vertebral body centres across 162 scans** and exported their 3D coordinates in millimetres. The target is the thick front part of each vertebra, shown in the "Body estimate" panels.

For each predicted vertebra, I kept the largest connected region and shrank it by 6 mm to break thin links to the rear parts of the bone. I kept the largest remaining core, grew it back by 6 mm within the original prediction, then trimmed behind the estimated back edge of the body. I checked body estimates visually on source-validation cases, without systematically optimizing the fixed 6 mm radius.

The centre is the average position of the retained voxels, converted to world RAS coordinates using the CT header. One JSON per scan records L1–L5 centres and flags missing predictions or failed separation.

Incomplete or mislabelled predictions can give unreliable centres. Centre accuracy was not measured against independent reference points.

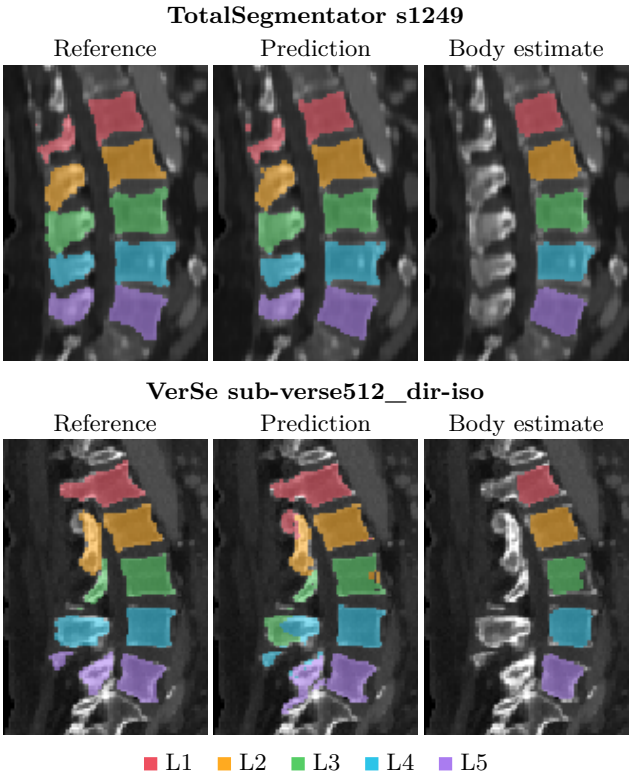


Figure 2. Frozen epoch-15 outputs on 2 mm RAS sagittal CT (superior up, anterior right). Both cases are near their dataset's median positive-case score. L1–L5 numbering agrees with the reference in these examples, although boundaries differ. Reference labels were used only for display and scoring.

D. Results and what they mean

Predictions were scored on each CT's native grid. Dice is mean ± sample SD across scans with an annotated target (n); misses score zero and absent levels are excluded. Macro Dice averages the five level means.

Level	TotalSegmentator Dice ± SD (n)	VerSe Dice ± SD (n)
L1	0.846 ± 0.238 (64)	0.775 ± 0.271 (58)
L2	0.884 ± 0.197 (59)	0.754 ± 0.307 (58)
L3	0.865 ± 0.241 (59)	0.699 ± 0.312 (58)
L4	0.894 ± 0.177 (60)	0.602 ± 0.394 (57)
L5	0.923 ± 0.038 (57)	0.597 ± 0.398 (51)
Macro Dice	**0.882**	**0.685**

Nine annotated source levels had zero overlap; **39/282 annotated VerSe levels had Dice below 0.10**. False lumbar predictions occurred in 1/19 entirely negative source scans and 3/15 negative VerSe scans.

VerSe performed worse, especially at L4/L5. Reviewed failures included a missed partial vertebra and a one-level numbering shift in sub-verse604. Differences in coverage, anatomy and scanners may contribute, but their effects were not isolated.

Limits and next steps. Earlier baseline test results had been inspected during development, although final checkpoint selection used source validation. With more resources, I would expand source training, check partial scans and level numbering on source validation, and validate body centres against independent references. I would avoid tuning on either test set because it would bias the evaluation.

Initial research

Before implementation, I used a customized deep research agent with the system prompt: "Act as a medical-imaging research engineer. Research the practical decisions needed for lumbar CT segmentation project before implementation. Research only. Do not launch training, purchase compute."